Kernelized Koopman Representation Geometry Governs Transient Amplification in Neural Dynamical Systems

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Abstract

We study how kernel representation geometry shapes finite-time behavior in the Koopman surrogates of neural dynamical systems. Although data-driven algorithms to reconstruct such transfer operators, in particular, Koopman operators, are common, their issue of producing correlation-based transient amplification remains largely unexplored. This phenomenon is due to the kernel-induced representation geometry on the Koopman operator from the corresponding reproducing kernel Hilbert space (RKHS) and is not apparent from the spectrum alone. We quantify this framework with the help of finite-time operator growth, pesudospectral analysis, and the Kreiss constant and apply them to three neural dynamical systems. Across these systems, different kernel representations produce markedly different finite-time amplification and pseudospectral behavior despite being trained on the same underlying dynamics. Our results demonstrate the existence of kernel hierarchy in four kernels for the Koopman operator representation inside respective RKHSs, and therefore those Koopman surrogates are not representation neutral. Our analysis can serve as a practical tool for assessing kernel Koopman models of neural dynamical systems before they are employed for prediction or perturbation-response analysis.

Keywords: Neural Dynamics, Transient Amplification, Kernel Representation Geometry

1. Introduction

The Koopman operator methods have been emerging as a popular data-driven modeling methods of neural dynamical systems (Sekizawa et al., 2026; Tang and Li, 2026; Cowan et al., 2016). The Koopman operator (Koopman, 1931; Brunton et al., 2022) recasts the evolution of nonlinear dynamics into linearly evolving observables. Due to the rich practical importance of this operator theoretic framework, research around compact representation of the Koopman operator has become abundant; for example (Schmid, 2010; Colbrook and Hansen, 2022; Colbrook, 2023; Colbrook et al., 2023; Lusch et al., 2018; Korda and Mezić, 2018; Williams et al., 2015a,b; Das et al., 2021). The finite representations of the Koopman operators is typically accompanied by its eigenvalues and eigenfunctions that essentially encodes the governing features (spatial and temporal) of dynamical systems. Therefore, to extract meaningful information, it is imperative to have a prior understanding of how these representations will cast the Koopman operators.

Of such compact representation methods, *Kernel Extended Dynamic Mode Decomposition* (KEDMD) (Williams et al., 2015b; Kawahara, 2016) provides a promising and attractive perspective because of their unique ability to capture dynamic measurements when the

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Table 1: Neural dynamical systems for kernel representation study. Refer to Appendix A for the mathematical equations and parameters of these systems.

System	Scale	Dynamical regime	Reference
Linear E–I	Population	Stable non-normal excitatory inhibitory dynamics	Controlled benchmark inspired by classical E–I population dynamics Wilson and Cowan (1972).
Wilson–Cowan	Population	Nonlinear excitatory–inhibitory dynamics near a stable equilibrium	Canonical neural-population model of Wilson and Cowan (1972).
FitzHugh–Nagumo	Single neuron	Stable fast–slow excitable dynamics	Reduced neuronal excitability model of FitzHugh (1961); Nagumo et al. (1962).

state space is huge. However, this attractive approach conceals a consequential modeling decision for the dynamical systems. Since every kernel has a unique correspondence to its native RKHS (Aronszajn, 1950; Wendland, 2005), therefore, the resulting approximation of the Koopman operators inside the respective RKHS is impacted by its function spaces and the geometry of the norm topology.

In such a setting, most literature, such as (Conradie et al., 2026; Köhne et al., 2025; Sinha et al., 2020; Kostic et al., 2023) and etc. inquires about superficial but important aspects such as whether the representation improves the prediction, operator reconstruction, invariant-subspace approximation, or spectral accuracy. In contrast to this, *our* paper discusses a subtle and deep concern when such kernelized Koopman representations are employed for the neural dynamical systems. These concerns are based on the finite-time operator growth when the kernelized Koopman representation is employed, which leads to the formation of a transient amplification phenomenon even when the spectrum of the operator appears benign. From an ML perspective, this indicates a profound notion of *representation quality*, where two representations should not be regarded as dynamically interchangeable simply because they describe the same training trajectories or possess similar spectra.

Contribution: 1) We show the transient amplification phenomena (Proposition 9) for the kernelized representation of the Koopman operator through KEDMD correlations in three standard neural dynamical systems (Section 4). 2) We show the existence of a kernel hierarchy with the use of four kernel functions. 3) We quantify this phenomenon with the use of pseudospectral analysis and the Kreiss constant (Theorem 10).

Problem Statement: After reviewing the necessary background in Section 2, we ask that, given the neural data, determine when the KEDMD surrogate (5) produces model correlations $C_f^{(\lambda)}(t) = v^* \mathbf{G}_\lambda A_\lambda^t v$ that exhibit transient amplification relative to the true correlations defined in (3). Then, considering the four standard kernels (Figure 1), characterize this instability in terms of kernel-induced geometry, induced operator growth, pseudospectral inflation, and the Kreiss constant of the normalized surrogate $\widetilde{A}_\lambda = \mathbf{G}_\lambda^{1/2} A_\lambda \mathbf{G}_\lambda^{-1/2}$.

Notation: We denote $\mathbb{N}/\mathbb{R}/\mathbb{C}$ as natural numbers, real numbers, and complex numbers, respectively. $L^2(\mu)$ is a Hilbert space with reference measure μ that contains square in-

tegrable functions defined over a Euclidean domain with the traditional inner product $\langle f, g\rangle_\mu = \int f(x)g(x)\mathrm{d}\mu(x)$. Unless otherwise specified, we have reserved t to signify the discrete time step value ($t \in \mathbb{N}$) and I as the *identity operator*. For a square matrix A, $\sigma(A)$ denotes its spectrum, $\sigma(A) \coloneqq \{z \in \mathbb{C} : \det(zI - A) = 0\}$, and, for $\varepsilon > 0$, its ε-pseudospectrum is $\sigma_\varepsilon(A) \coloneqq \{z \in \mathbb{C} : \|(zI - A)^{-1}\|_2 > \varepsilon^{-1}\}$.

2. Kernel Representation Geometry for Koopman Dynamics

2.1. The Koopman operator for Dynamical Systems

Let $\Omega \subset \mathbb{R}^d$ be an open domain and consider $\boldsymbol{F} : \Omega \to \mathbb{R}^d$ as follows:

$$x_{n+1} = \boldsymbol{F}(x_n) \tag{1}$$

with $n \in \mathbb{N} \cup \{0\}$. The discrete dynamical system in (1) can be inferred from ODE $\dfrac{\mathrm{d}}{\mathrm{d}t}x(t) = \boldsymbol{f}(x(t))$ where $x(t) \in \Omega$ represents the dynamical state at time $t \geq 0$ and $\boldsymbol{f} : \Omega \to \mathbb{R}^d$ is a locally Lipschitz map. The dynamic flow $\boldsymbol{F}$ naturally induces the associated linear Koopman operator defined as follows.

Definition 1 *Let $(\Omega, \mathcal{B}, \mu)$ be a measurable space, and let $\boldsymbol{F} : \Omega \to \mathbb{R}^d$ as defined in (1). The Koopman operator denoted by U with the underlying dynamic symbol $\boldsymbol{F}$ acts on smooth* observables $f : \Omega \to \mathbb{C}$ *by* composition *with $\boldsymbol{F}$,*

$$(Uf)(x) \coloneqq f \circ \boldsymbol{F}(x). \tag{2}$$

An observable $\varphi : \Omega \to \mathbb{C}$ is called an *eigenfunction* of the Koopman operator U if it satisfies $U\varphi = \xi\varphi$ where $\xi \in \mathbb{C}$ is the eigenvalue of U. When μ has invariant probability distributions (*measure preserving* (Lasota and Mackey, 2013)), i.e. $\mu(\boldsymbol{F}^{-1}(B)) = \mu(B)$ for all $B \in \mathcal{B}$, then $U : L^2(\mu) \to L^2(\mu)$ is a *unitary* operator, and then $\xi \in \mathbb{T} \coloneqq \{z \in \mathbb{C} : |z| = 1\}$. If we have an observable $g \in \mathsf{span}\{\varphi_i \mid i \in \mathbb{N}\}$ where φ_i are the Koopman eigenfunctions, there exist corresponding Koopman modes $\wp_i \in \mathbb{C}$ of the observable g, then we have the following *Koopman mode decomposition* (Mezić, 2005):

$$(Ug)(x_t) = \sum_{i \in \mathbb{N}} \xi_i^t \wp_i \varphi_i(x_0).$$

The Koopman mode decomposition formula given above captures the governing features of the dynamics $\boldsymbol{F}$. In particular, it decomposes the expected dynamics $\boldsymbol{F}$ observed by the observable g into *complex stationary modes* $\wp_i$, where the eigenvalues ξ_i dictate the temporal measurements and the eigenfunctions φ_i dictate the spatial measurements.

The Koopman operator may not be a normal compact operator, and hence its eigenfunctions may fail to form a complete orthonormal basis of the space, thereby making the Koopman operator learning challenging. We therefore consider the time-lagged observable *correlation* (Lasota and Mackey, 2013) defined as follows for the Koopman operator. The correlation in (3) can be used to define the notion of numerical radius. For a bounded linear operator $\mathscr{T}$ on a Hilbert space $\mathcal{H}$, its *numerical radius* is $\boldsymbol{w}(\mathscr{T}) \coloneqq \sup_{f \in \mathcal{H}\setminus\{0\}} \dfrac{|\langle f, \mathscr{T}f\rangle_{\mathcal{H}}|}{\|f\|_{\mathcal{H}}^2}$.

Definition 2 (Correlation) *For an observable $f : \Omega \to \mathbb{C}$ such that $f \in L^2(\mu)$, the time-t correlation with itself is*

$$C_f(t) := \langle f, U^t f \rangle_\mu = \mathbb{E}_{x_0 \sim \mu} \left[\overline{f(x_0)} f(x_t) \right], \qquad x_t = \boldsymbol{F}^t(x_0) \tag{3}$$

where $U^t f = f \circ \boldsymbol{F}^t$ and $\boldsymbol{F}^t$ is the $t-$fold composition of $\boldsymbol{F}$ with itself where $\boldsymbol{F}^1 \equiv \boldsymbol{F}$. Whenever U^t is bounded, $\boldsymbol{w}(U^t) = \sup_{f \neq 0} \dfrac{|C_f(t)|}{C_f(0)}$.

Definition 3 (Correlation-based Transient amplification) *Let $\mathscr{T}$ be a bounded linear operator on a Hilbert space $\mathcal{H}$. The finite-time correlation gain at time t is $\Gamma_{\mathscr{T}}(t) := \boldsymbol{w}(\mathscr{T}^t)$. We say that $\mathscr{T}$ exhibits correlation amplification at time t if $\Gamma_{\mathscr{T}}(t) > 1$. If the spectral radius of $\mathscr{T}$, $\rho(\mathscr{T}) < 1$, then this is called* correlation based transient amplification.

2.2. Kernelized Representation of the Koopman operator

Denote $\mathcal{X} := \{x_i\}_{i=1}^M \subset \mathbb{R}^d$ where $M \in \mathbb{N}$, as distinct data points sampled through (1) and let $\mathcal{Y} := \{\boldsymbol{F}(x_i)\}_{i=1}^M \subset \mathbb{R}^d$. Let $k : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$ be a continuous strictly positive definite symmetric kernel (denoted by $\succ 0$). At some arbitrary vector $x \in \mathbb{R}^d$, define the canonical feature $\phi_z(x) = k(x, z)$. Then, the linear space given by $\mathsf{span}\{\phi_z : z \in \mathbb{R}^d\}$ extends by completion to a Hilbert space $\mathcal{H}_k$, called the *reproducing kernel Hilbert space* (RKHS) generated by $k(\cdot, \cdot)$. For all functions $f \in \mathcal{H}_k$, the following properties hold true at $x \in \mathbb{R}^d$:

$$\langle f, k(\cdot, x) \rangle_{\mathcal{H}_k} = f(x) \tag{4a}$$

$$\langle k(\cdot, y), k(\cdot, x) \rangle_{\mathcal{H}_k} = k(x, y) \tag{4b}$$

$$|f(x)| \leq \|f\|_{\mathcal{H}_k} \cdot (k(x, x))^{1/2}. \tag{4c}$$

The property shown in (4a) is called the *reproducing property* of $k(\cdot, x)$; when $f = k(\cdot, y)$ in (4a), we get (4b) and finally (4c) shows the boundedness of $f \in \mathcal{H}_k$ in $x \in \mathbb{R}^d$. In the setting of the RKHS $\mathcal{H}_k$, it is imperative to associate the domain of the Koopman operators as $\mathrm{dom}(U) \subseteq \mathcal{H}_k$ with $(Uf)(x) = f \circ \boldsymbol{F}(x)$ where $f \in \mathcal{H}_k$. With an additional assumption that the Koopman operator U is *densely defined* (Pedersen, 2012) over $\mathcal{H}_k$, that is, $\overline{\mathrm{dom}(U)} = \mathcal{H}_k$, then one can have a well-defined *adjoint* of the Koopman operator U^* acting on the kernel sections $k_x := k(\cdot, x)$; Lemma 4 is the result that describes the same.

Lemma 4 *Consider the Koopman operator $U : \mathrm{dom}(U) \subseteq \mathcal{H}_k \to \mathcal{H}_k$, $Uf = f \circ \boldsymbol{F}$ where $\boldsymbol{F}$ is defined in (1). If $\overline{\mathrm{dom}(U)} = \mathcal{H}_k$, then $k_x \in \mathrm{dom}(U_F^*)$ and $U^* k_x = k_{\boldsymbol{F}(x)}$.*

KEDMD provides a finite dimensional approximate surrogate for the Koopman operator (2) with the use of the kernels in their corresponding RKHS. Recall the sampled dataset $\mathcal{X}$ and define $V_{\mathcal{X}} := \mathsf{span}\{k(\cdot, x_i) : x_i \in \mathcal{X},\ i = 1, \ldots, M\}$. Consider an orthogonal projection $P_{\mathcal{X}}$ from $\mathcal{H}_k$ to $V_{\mathcal{X}}$, i.e. $P_{\mathcal{X}} f$ solves the regression problem: $\min_{g \in V_{\mathcal{X}}} \|f - g\|_{\mathcal{H}_k}$. Then, (Köhne et al., 2025, Proposition 3.2), a finite dimensional KEDMD approximant of the Koopman operator U in RKHS $\mathcal{H}_k$ is $\widehat{U} = P_{\mathcal{X}} U$ and the same is explicitly given as:

$$\widehat{U} f = P_{\mathcal{X}} (Uf) = P_{\mathcal{X}} (f \circ \boldsymbol{F}) = \mathbf{G}_{\mathcal{X}}^{-1} \mathbf{A}_{\mathcal{X}},$$

Kernel Representations

Kernel	$k(x, z)$
Laplace	$\exp(-\|x - z\|_1/\gamma)$
Matérn-3/2	$\left(1 + \frac{\sqrt{3}r}{\gamma}\right)\exp(-\sqrt{3}r/\gamma)$
RQ	$\left(1 + \frac{r^2}{2\gamma^2}\right)^{-1}$
Gaussian RBF	$\exp\left(-r^2/(2\gamma^2)\right)$

$r = \|x - z\|_2$, except that the Laplace kernel uses $\|x - z\|_1$, and $\gamma > 0$ denotes the kernel scale.

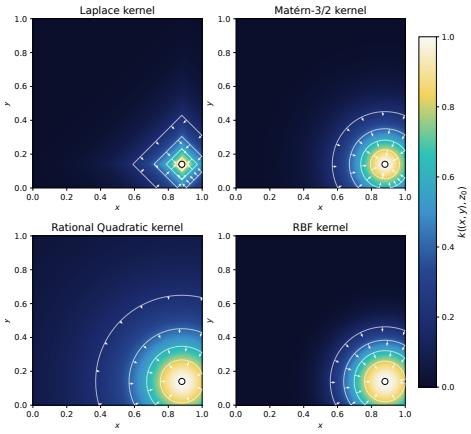


Figure 1: **Kernel representations and their induced local geometries.** Left: analytic forms of the four kernels. Right: Representative kernel sections $k((x, y), z_0)$ on $[0, 1]^2$ for $z_0 = (0.88, 0.14)$, illustrating the distinct norm geometries induced by the four kernels. White contours show equal-similarity level sets, while arrows indicate the local direction of increasing kernel similarity, $+\nabla_{(x,y)}k((x, y), z_0)$.

where the *interaction matrix* $\mathbf{A}_{\mathcal{X}} \coloneqq [k(x_{i+1}, x_j)]_{i,j=1}^{M} \in \mathbb{R}^{M\times M}$ and the *kernel Gram matrix* $\mathbf{G}_{\mathcal{X}} \coloneqq [k(x_i, x_j)]_{i,j=1}^{M} \in \mathbb{R}^{M\times M}$. We also systematically incorporate the RKHS feature operator $\mathbf{\Phi}_{\mathcal{X}} : \mathbb{R}^M \to \mathcal{H}_k$ defined by $\mathbf{\Phi}_{\mathcal{X}} \coloneqq \begin{bmatrix} k_{x_1} & \cdots & k_{x_M} \end{bmatrix}$ and therefore we get $\mathbf{G}_{\mathcal{X}} = \mathbf{\Phi}_{\mathcal{X}}^*\mathbf{\Phi}_{\mathcal{X}}$ and $\mathbf{A}_{\mathcal{X}} = \mathbf{\Phi}_{\mathcal{Y}}^*\mathbf{\Phi}_{\mathcal{X}}$. We provide the Tikhonov regularization to the Koopman surrogate $\widehat{U}f$ by considering regularizer $0 < \lambda \ll 1$ and let $\mathbf{G}_\lambda \coloneqq (\mathbf{G}_{\mathcal{X}} + \lambda I)$:

$$A_\lambda = (\mathbf{G}_{\mathcal{X}} + \lambda I)^{-1}\mathbf{A}_{\mathcal{X}} = \mathbf{G}_\lambda^{-1}\mathbf{A}_{\mathcal{X}} = (\mathbf{\Phi}_{\mathcal{X}}^*\mathbf{\Phi}_{\mathcal{X}} + \lambda I)^{-1}\left(\mathbf{\Phi}_{\mathcal{Y}}^*\mathbf{\Phi}_{\mathcal{X}}\right). \tag{5}$$

Figure 1 tabulates the standard positive-definite real bounded kernels (Rasmussen, Carl Edward and Williams, Christopher KI and others, 2006) used in our study of kernel representation for neural dynamical systems. We also let $\mathbf{G}_{\mathcal{Y}} \coloneqq [k(\boldsymbol{F}(x_i), \boldsymbol{F}(x_j))]_{i,j=1}^{M} \in \mathbb{R}^{M\times M}$.

Let $V_{\mathcal{Y}} \coloneqq \mathsf{span}\{k_{\boldsymbol{F}(x_1)}, \ldots, k_{\boldsymbol{F}(x_M)}\}$. The following Proposition 5 quantifies the deformation of the sampled kernel-induced geometry under the same physical dynamics $\boldsymbol{F}$ through the restriction of the adjoint of the Koopman operator U from the Lemma 4.

Proposition 5 *Define* $T_{\boldsymbol{F},\mathcal{X}} \coloneqq U^*\big|_{V_{\mathcal{X}}}$ *by* $T_{\boldsymbol{F},\mathcal{X}} : V_{\mathcal{X}} \to V_{\mathcal{Y}}$ *by* $T_{\boldsymbol{F},\mathcal{X}}\left(\sum_{i=1}^{M} c_i k_{x_i}\right) = \sum_{i=1}^{M} c_i k_{\boldsymbol{F}(x_i)}$ *where* $\boldsymbol{c} \coloneqq \begin{bmatrix} c_1 & \cdots & c_M \end{bmatrix}$ *is the coefficient vector. Then* $\|T_{\boldsymbol{F},\mathcal{X}}\|^2_{\mathcal{H}_k\to\mathcal{H}_k} = \lambda_{\max}\left(\mathbf{G}_{\mathcal{X}}^{-1/2}\mathbf{G}_{\mathcal{Y}}\mathbf{G}_{\mathcal{X}}^{-1/2}\right)$, *where* $\lambda_{\max}(\bullet)$ *is the maximum eigenvalue of* $\bullet$.

The following definition measures the non-normality of a matrix $\mathscr{A}$ by quantifying the resolvent amplification outside the unit disk (Kreiss, 1962), which will be used in our study in the latter part to investigate pseudospectral analysis of the KEDMD surrogates.

Definition 6 *The Kreiss constant of a matrix* $\mathscr{A}$ *is*

$$\mathcal{K}(\mathscr{A}) \coloneqq \sup_{|z|>1}\left\{(|z| - 1)\cdot\|(zI - \mathscr{A})^{-1}\|\right\}. \tag{6}$$

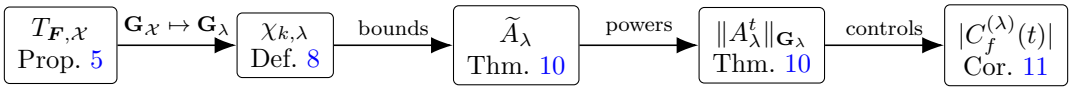


Figure 2: Logical progression from finite kernel-section transport $T_{\boldsymbol{F},\mathcal{X}} \coloneqq U^{*}\big|_{V_{\mathcal{X}}}$ to observable finite-time response $|C_{f}^{(\lambda)}(t)|$.

3. Transient Amplification and Pseudospectral Diagnostics

In this section, we provide the theoretical outlook for the transient amplification and pseudospectral analysis that exist in the surrogate of the kernelized representation of the Koopman operator and are summarized in Figure 2. Recall the KEDMD surrogate from (5).

Definition 7 *For a complex vector $v \in \mathbb{C}^{M}$, define the $\mathbf{G}_{\lambda}$-geometry norm by $\|v\|_{\mathbf{G}_{\lambda}}^{2} \coloneqq v^{*}\mathbf{G}_{\lambda}v$, and the induced operator norm by $\|A\|_{\mathbf{G}_{\lambda}} \coloneqq \sup_{v\neq 0} \dfrac{\|Av\|_{\mathbf{G}_{\lambda}}}{\|v\|_{\mathbf{G}_{\lambda}}}$.*

The following construction in Definition 8 separates three levels of kernel representation geometry. The kernel Gram matrix $\mathbf{G}_{\mathcal{X}}$ encodes the geometry of the sampled kernel sections before evolution, while $\mathbf{G}_{\mathcal{Y}}$ records that geometry after the same physical map $\boldsymbol{F}$ is applied. Their relative deformation is measured by finite kernel-section transport, and its regularized form $\chi_{k,\lambda}(\boldsymbol{F};\mathcal{X})$ enters the normalized KEDMD surrogate.

Definition 8 *Define regularized transport as $\chi_{k,\lambda}(\boldsymbol{F};\mathcal{X}) \coloneqq \left\|\mathbf{G}_{\lambda}^{-1/2}\mathbf{G}_{\mathcal{Y}}\mathbf{G}_{\lambda}^{-1/2}\right\|_{2}^{1/2}$. Furthermore, $\chi_{k,0}(\boldsymbol{F};\mathcal{X}) = \|T_{\boldsymbol{F},\mathcal{X}}\|$.*

Proposition 9 *Let the KEDMD correlation be defined as $C_{f}^{(\lambda)}(t) = v^{*}\mathbf{G}_{\lambda}A_{\lambda}^{t}v$. Then $|C_{f}^{(\lambda)}(t)| \leq \|v\|_{\mathbf{G}_{\lambda}}^{2}\,\|A_{\lambda}^{t}\|_{\mathbf{G}_{\lambda}}$ holds for $t \geq 0$. If $\|A_{\lambda}\|_{\mathbf{G}_{\lambda}} \geq 1$, then $|C_{f}^{(\lambda)}(t)| \leq \|v\|_{\mathbf{G}_{\lambda}}^{2}\,\left\|A_{\lambda}^{t}\right\|_{\mathbf{G}_{\lambda}}$.*

The above proposition isolates three key ingredients that govern the correlation based transient amplification of the KEDMD surrogate, which are the observable coefficient size through the vector v, the induced growth of the operator through $\|A_{\lambda}\|_{\mathbf{G}_{\lambda}}$ and the kernel Gram conditioning.

Theorem 10 *Consider: $\widetilde{A}_{\lambda} \coloneqq \mathbf{G}_{\lambda}^{1/2}A_{\lambda}\mathbf{G}_{\lambda}^{-1/2} = \mathbf{G}_{\lambda}^{-1/2}\mathbf{A}_{\mathcal{X}}\mathbf{G}_{\lambda}^{-1/2}$, where $A_{\lambda} = \mathbf{G}_{\lambda}^{-1}\mathbf{A}_{\mathcal{X}}$ and $\mathbf{G}_{\lambda}$ is positive definite. Then, the following holds for every $t \geq 0$,*

1. $\|\tilde{A}_{\lambda}\|_{2} \leq \chi_{k,\lambda}(\boldsymbol{F};\mathcal{X})$.
2. $\left\|A_{\lambda}^{t}\right\|_{\mathbf{G}_{\lambda}} = \left\|\widetilde{A}_{\lambda}^{t}\right\|_{2}$.
3. $\left\|A_{\lambda}^{t}\right\|_{\mathbf{G}_{\lambda}} = \left\|\widetilde{A}_{\lambda}^{t}\right\|_{2} \leq \chi_{k,\lambda}(\boldsymbol{F};\mathcal{X})^{t}$.
4. *If $\widetilde{A}_{\lambda}$ is normal, then $\left\|A_{\lambda}^{t}\right\|_{\mathbf{G}_{\lambda}} = \left\|\widetilde{A}_{\lambda}^{t}\right\|_{2} = \left(\rho(\widetilde{A}_{\lambda})\right)^{t}$, where $\rho(\widetilde{A}_{\lambda})$ denotes the spectral radius of $\widetilde{A}_{\lambda}$.*

5. *If* $\sigma(\widetilde{A}_\lambda) \subseteq \{z : |z| \le 1\}$, *then* $\mathcal{K}(\widetilde{A}_\lambda) \le \sup_t \|A_\lambda^t\|_{\mathbf{G}_\lambda} \le \mathrm{e} \cdot n \cdot \mathcal{K}(\widetilde{A}_\lambda)$, *where* $n \in \mathbb{N}$ *is the KEDMD surrogate dimension. Furthermore, if* $\sigma_\varepsilon(\widetilde{A}_\lambda)$ *extends to* $r_\varepsilon(\widetilde{A}_\lambda)$*:*

$$r_\varepsilon(\widetilde{A}_\lambda) := \sup\{|z| : z \in \sigma_\varepsilon(\widetilde{A}_\lambda)\}. \tag{7}$$

Then $\mathcal{K}(\widetilde{A}_\lambda) \ge \varepsilon^{-1}(r_\varepsilon(\widetilde{A}_\lambda) - 1)$. *Therefore,* $\sup_{t\ge 0} \left\|A_\lambda^t\right\|_{\mathbf{G}_\lambda} \ge \varepsilon^{-1}(r_\varepsilon(\widetilde{A}_\lambda) - 1)$.

The key aspects of Theorem 10 establish that transient amplification in the KEDMD surrogate is fundamentally controlled by the non-normal geometry of the kernel-induced representation rather than by eigenvalues alone. The $\mathbf{G}_\lambda$-induced dynamics is exactly equivalent to the Euclidean dynamics of the normalized operator $\tilde{A}_\lambda$. When this operator is normal, its spectral radius completely determines finite-time growth, whereas non-normality permits substantial transient amplification despite apparently stable eigenvalues. Lastly, the Kreiss and pseudospectral bounds provide a spectral analysis of this amplification, connecting outward pseudospectral expansion to unavoidable finite-time growth.

The following corollary identifies the observable-level consequence of the preceding operator theory in terms of the numerical radius of $\tilde{A}_\lambda$ and $\chi_{k,\lambda}(\boldsymbol{F}; \mathcal{X})$.

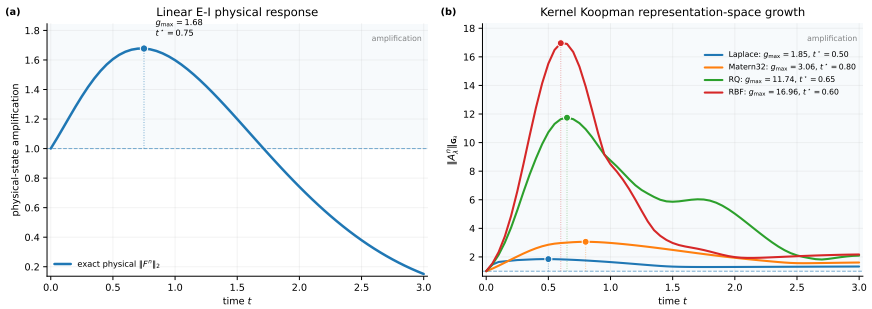
Corollary 11 *Recall* $C_f^{(\lambda)}(t) := v^* \mathbf{G}_\lambda A_\lambda^t v$ *from Proposition 9 and* $C_f^{\lambda}(0) = v^* \mathbf{G}_\lambda v = \|v\|_{\mathbf{G}_\lambda}^2$. *Let* $\boldsymbol{w}(\tilde{A}_\lambda^t) = \sup_{v \in \mathbb{C}^M \setminus \{0\}} \dfrac{|C_f^{(\lambda)}(t)|}{C_f^{(\lambda)}(0)}$ *be the numerical radius of* $\tilde{A}_\lambda^t$. *Then:*

$$\boldsymbol{w}(\tilde{A}_\lambda^t) \le \left\|\tilde{A}_\lambda^t\right\|_2 = \left\|A_\lambda^t\right\|_{\mathbf{G}_\lambda} \le \chi_{k,\lambda}(\boldsymbol{F}; \mathcal{X})^t.$$

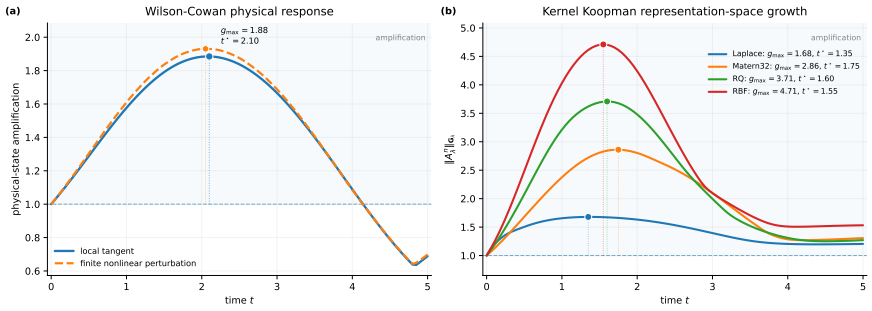
4. Experiments

4.1. Transient Amplification

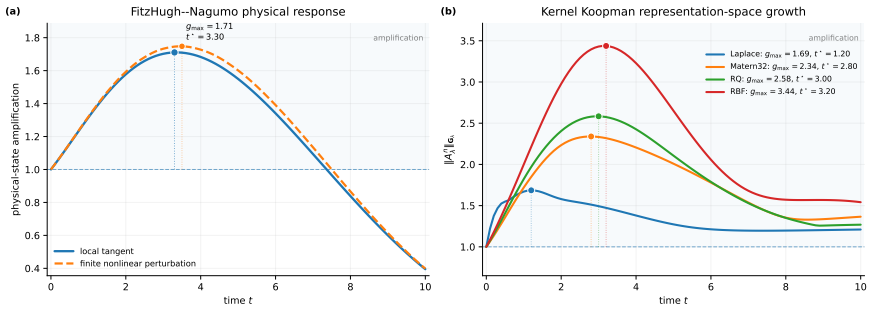
Figure 3 shows the correlation based transient amplification of the neural dynamical system considered in Table 1. For the linear E–I benchmark, the exact physical propagator exhibits only moderate transient growth, with a peak amplification of ~ 1.68. In contrast, the corresponding KEDMD surrogates exhibit markedly different growth in representation-space, ranging from 1.85 for the Laplace kernel to 16.96 for the RBF kernel. Since all surrogates are constructed from the same underlying snapshot pairs, this variation cannot be attributed to changes in the physical dynamics but instead reflects differences induced by the kernel representation. The same qualitative behavior persists for the nonlinear Wilson–Cowan and FitzHugh–Nagumo neural dynamics. In both cases, the local tangent response closely agrees with the finite-amplitude nonlinear perturbation response over the regime considered, while the learned Koopman surrogates exhibit substantially kernel-dependent transient growth. Wilson–Cowan (Figure 3(b)) produces peak representation-space amplifications of approximately 1.68, 2.86, 3.71, and 4.71, while FitzHugh–Nagumo (Figure 3(c)) produces 1.69, 2.34, 2.58 and 3.44 for Laplace, Matérn-3/2, RQ, and Gaussian RBF kernels, respectively.



(a) Linear E-I



(b) Wilson-Cowan



(c) FitzHugh-Nagumo

Figure 3: **Representation dependent transient amplification across neural dynamical systems.** Physical-state responses (*left* within each pair) are compared with $\mathbf{G}_\lambda$-induced finite-time amplification of KEDMD surrogates (*right*). In this figure and Table 2 included, $g_{\max} := \max_n \|A_\lambda^n\|_{\mathbf{G}_\lambda}$ and $t^\star$ denotes the physical time at which $g_{\max}$ is attained. Different kernel representations of the same underlying dynamics exhibit markedly different transient growth, *with a* ***consistent Laplace–Matérn32–RQ–Gaussian RBF*** *amplification hierarchy.*

4.2. Pseudospectral and Kreiss Diagnostics

The kernel representation-dependent transient growth in Figure 3 is accompanied by the corresponding changes in the resolvent geometry of the normalized KEDMD surrogates.

Although their spectral radii remain close to unity, the pseudospectra differ systematically across kernel representations, revealing finite-time sensitivity that is not apparent from eigenvalues alone. Complete pseudospectral maps for all kernel–system combinations (total *12* combinations) are provided in Figure 6 in Appendix E. We summarize this resolvent sensitivity using the Kreiss estimate based on grids $\widehat{\mathcal{K}}$ in Table 2. The separation is noticeable for the linear E–I benchmark, where $\widehat{\mathcal{K}}$ increases from 1.371 for the Laplace to 5.773 for the Gaussian RBF. Thus, while the underlying neural system determines the magnitude of the resolvent response, the kernel representation consistently determines its ordering.

Table 2: **Representation-dependent finite-time and pseudospectral diagnostics across the three neural dynamical systems.** Here, $\widehat{\mathcal{K}}$ is the grid-based discrete Kreiss estimate, and $r_{10^{-2}}$ is the radial extent of the 10^{-2}-pseudospectrum.

Neural Dynamical System	**Kernel**	$\rho(\widetilde{A}_\lambda)$	$g_{\max}$	$t_\star$	$\widehat{\mathcal{K}}$	$r_{10^{-2}}$
Linear E–I	Laplace	1.0000	1.847	0.50	1.370	0.997
	Matérn-3/2	1.0000	3.058	0.80	1.676	1.000
	RQ	1.0000	11.742	0.65	4.261	1.014
	RBF	1.0000	**16.959**	0.60	**5.773**	1.044
Wilson–Cowan	Laplace	0.9999	1.679	1.35	1.251	0.998
	Matérn-3/2	1.0000	2.860	1.75	1.372	1.000
	RQ	1.0000	3.708	1.60	1.658	1.000
	RBF	1.0001	**4.710**	1.55	**2.091**	1.000
FitzHugh–Nagumo	Laplace	1.0000	1.686	1.20	1.200	0.997
	Matérn-3/2	1.0004	2.339	2.80	1.391	0.998
	RQ	1.0003	2.584	3.00	1.486	0.998
	RBF	1.0006	**3.437**	3.20	**1.796**	0.998

5. Conclusion

We studied the kernelized Koopman representations for the neural dynamical systems where we demonstrate that a learned Koopman surrogate is not determined by the underlying neural flow and snapshot data alone, rather the RKHS in which that dynamics is represented, is itself a consequential part of the finite-time dynamics of the surrogate. This distinction is invisible if the learned Koopman operator is studied only by its spectrum. Across stable E–I, Wilson–Cowan, and FitzHugh–Nagumo dynamics, we found that different kernel representations of the same snapshot evolution produce substantially different finite-time operator growth and Koopman resolvent geometry. With the standard choices of four kernels, we observe a consistent kernel hierarchy (see Figure 3) in the realized transient and Kreiss diagnostics (see Table 2).

Our research work also has certain limitations. In particular, our work quantifies the kernel representation geometry induced on sampled RKHSs sections and on their regularized KEDMD surrogates; however, our research does not yet completely characterize the Koopman composition operator on the respective RKHSs. Furthermore, $\chi_{k,\lambda}$ provides a one-step

geometric control rather than a sharp characterization of multi-step transient growth; the latter also depends on the nonnormal representation of the normalized surrogate.

Our prospective future directions include extending the investigation of transient amplification phenomena in complex neural dynamical systems. Taking inspiration from (Gonzalez et al., 2024), it is also important to investigate for a given neural flow, when the Koopman operators are densely defined, closable, bounded, or necessarily unbounded, and to determine how these infinite-dimensional properties are reflected in the finite transport operators $T_{\boldsymbol{F},\mathcal{X}}$. Understanding this operator theoretic relation would provide a principled route from sampled RKHS geometry to intrinsic properties of the Koopman operator itself.

References

Nachman Aronszajn. Theory of reproducing kernels. *Transactions of the American Mathematical Society*, 68(3):337–404, 1950. URL https://doi.org/10.1090/S0002-9947-1950-0051437-7.

Steven L. Brunton, Marko Budišić, Eurika Kaiser, and J. Nathan Kutz. Modern Koopman Theory for Dynamical Systems. *SIAM Rev.*, 64(2):229–340, January 2022. ISSN 0036-1445. doi: 10.1137/21M1401243. URL https://doi.org/10.1137/21M1401243.

Matthew J Colbrook. The `mpEDMD` Algorithm for Data-Driven Computations of Measure-Preserving Dynamical Systems. *SIAM Journal on Numerical Analysis*, 61(3):1585–1608, 2023.

Matthew J Colbrook and Anders C Hansen. The foundations of spectral computations via the Solvability Complexity Index hierarchy. *Journal of the European Mathematical Society*, 25(12):4639–4718, 2022.

Matthew J Colbrook, Lorna J Ayton, and Máté Szőke. Residual Dynamic Mode Decomposition: Robust & verified Koopmanism. *Journal of Fluid Mechanics*, 955:A21, 2023.

Gustav Conradie, Nicolas Boullé, Jean-Christophe Loiseau, Steven L Brunton, and Matthew J Colbrook. Trustworthy Koopman Operator Learning: Invariance Diagnostics and Error Bounds. *arXiv preprint arXiv:2603.15091*, 2026.

Jack D Cowan, Jeremy Neuman, and Wim van Drongelen. Wilson–Cowan Equations for Neocortical Dynamics. *The Journal of Mathematical Neuroscience*, 6(1):1, 2016.

Suddhasattwa Das, Dimitrios Giannakis, and Joanna Slawinska. Reproducing kernel Hilbert space compactification of unitary evolution groups. *Applied and Computational Harmonic Analysis*, 54:75–136, 2021.

Richard FitzHugh. Impulses and Physiological States in Theoretical Models of Nerve Membrane. *Biophysical journal*, 1(6):445–466, 1961.

Efrain Gonzalez, Moad Abudia, Michael Jury, Rushikesh Kamalapurkar, and Joel A Rosenfeld. The kernel perspective on dynamic mode decomposition. *Transactions on Machine Learning Research*, 2024. ISSN 2835-8856. URL https://openreview.net/forum?id=sIR8xV7hGl.

Karl E Gustafson and Duggirala KM Rao. Numerical Range. In *Numerical Range: The Field of Values of Linear Operators and Matrices*, pages 1–26. Springer, 1997. doi: https://doi.org/10.1007/978-1-4613-8498-4.

Yoshinobu Kawahara. Dynamic mode decomposition with reproducing kernels for koopman spectral analysis. *Advances in Neural Information Processing Systems*, 29, 2016.

Frederik Köhne, Friedrich M. Philipp, Manuel Schaller, Anton Schiela, and Karl Worthmann. $\boldsymbol{L}^{\infty}$-error bounds for approximations of the koopman operator by kernel extended dynamic mode decomposition. *SIAM Journal on Applied Dynamical Systems*, 24(1):501–529, 2025. doi: 10.1137/24M1650120. URL https://doi.org/10.1137/24M1650120.

Bernard O Koopman. Hamiltonian systems and transformation in Hilbert space. *Proceedings of the National Academy of Sciences*, 17(5):315–318, 1931. URL https://www.pnas.org/doi/pdf/10.1073/pnas.17.5.315.

Milan Korda and Igor Mezić. On Convergence of Extended Dynamic Mode Decomposition to the Koopman Operator. *Journal of Nonlinear Science*, 28(2):687–710, 2018. URL https://link.springer.com/content/pdf/10.1007/s00332-017-9423-0.pdf.

Vladimir Kostic, Karim Lounici, Pietro Novelli, and Massimiliano Pontil. Sharp spectral rates for Koopman operator learning. *Advances in Neural Information Processing Systems*, 36:32328–32339, 2023.

Heinz-Otto Kreiss. Über die stabilitätsdefinition für differenzengleichungen die partielle differentialgleichungen approximieren. *BIT Numerical Mathematics*, 2(3):153–181, 1962.

Andrzej Lasota and Michael C Mackey. *Chaos, fractals, and noise: stochastic aspects of dynamics*, volume 97. Springer Science & Business Media, 2013.

Bethany Lusch, J Nathan Kutz, and Steven L Brunton. Deep learning for universal linear embeddings of nonlinear dynamics. *Nature communications*, 9(1):4950, 2018. URL www.nature.com/articles/s41467-018-07210-0.pdf.

Igor Mezić. Spectral properties of dynamical systems, model reduction and decompositions. *Nonlinear Dynamics*, 41:309–325, 2005.

Jinichi Nagumo, Suguru Arimoto, and Shuji Yoshizawa. An active pulse transmission line simulating nerve axon. *Proceedings of the IRE*, 50(10):2061–2070, 1962.

Gert K Pedersen. *Analysis now*, volume 118. Springer Science & Business Media, 2012.

Rasmussen, Carl Edward and Williams, Christopher KI and others. *Gaussian Processes for Machine Learning*, volume 1. Springer, 2006.

Peter J Schmid. Dynamic Mode Decomposition of Numerical and Experimental Data. *Journal of Fluid Mechanics*, 656:5–28, 2010.

Daiki Sekizawa, Sosuke Ito, and Masafumi Oizumi. Koopman mode decomposition of thermodynamic dissipation in nonlinear Langevin dynamics. *Proceedings of the National Academy of Sciences*, 123(25):e2530617123, 2026.

Subhrajit Sinha, Bowen Huang, and Umesh Vaidya. On Robust Computation of Koopman Operator and Prediction in Random Dynamical Systems. *Journal of Nonlinear Science*, 30(5):2057–2090, 2020.

XuJiang Tang and QiongLin Li. Structure-preserving Koopman predictive control for memristive neural dynamics: input-exact and commutator-defect lifting. *Biological Cybernetics*, 120(5):26, 2026.

Lloyd N Trefethen and Mark Embree. Spectra and pseudospectra: the behavior of nonnormal matrices and operators. 2020.

Holger Wendland. *Scattered Data Approximation*, volume 17. Cambridge University Press Cambridge, 2005.

Matthew O Williams, Ioannis G Kevrekidis, and Clarence W Rowley. A data–driven approximation of the Koopman operator: Extending Dynamic Mode Decomposition. *Journal of Nonlinear Science*, 25:1307–1346, 2015a.

Matthew O. Williams, Clarence W. Rowley, and Ioannis G. Kevrekidis. A kernel-based method for data-driven Koopman spectral analysis. *Journal of Computational Dynamics*, 2(2):247–265, 2015b. ISSN 2158-2491. doi: 10.3934/jcd.2015005. URL https://www.aimsciences.org/article/id/ce535396-f8fe-4aa1-b6de-baaf986f6193.

Hugh R Wilson and Jack D Cowan. Excitatory and inhibitory interactions in localized populations of model neurons. *Biophysical journal*, 12(1):1–24, 1972.

Appendix A. Neural Dynamical Systems

A.1. Linearized Excitatory–Inhibitory Dynamics

We first consider a stable linearized excitatory–inhibitory (E–I) population model,

$$\dot{\mathbf{x}}(t) = J\mathbf{x}(t), \qquad \mathbf{x}(t) = \begin{bmatrix} \delta E(t) \\ \delta I(t) \end{bmatrix},$$

where δE and δI denote deviations of the excitatory and inhibitory population activities from equilibrium. We use

$$J = \begin{bmatrix} 0.8 & -4.0 \\ 1.0 & -2.8 \end{bmatrix} \in \mathbb{R}^{2\times 2}.$$

For a sampling interval Δt, the exact discrete-time flow is

$$\mathbf{x}_{n+1} = F_{\Delta t}\mathbf{x}_n, \qquad F_{\Delta t} = \exp(J\Delta t),$$

with $\Delta t = 0.05$ in our experiments.

A.2. Wilson–Cowan Population Dynamics

We consider the Wilson–Cowan system from (Wilson and Cowan, 1972):

$$\tau_E \dot{E} = -E + (1 - rE)S(w_{EE}E - w_{EI}I + P)\,,$$
$$\tau_I \dot{I} = -I + (1 - rI)S(w_{IE}E - w_{II}I + Q)\,,$$

where $E(t)$ and $I(t)$ denote the excitatory and inhibitory population activities, respectively. The nonlinear activation is given by the sigmoidal function as follows:

$$S(u) = \frac{1}{1 + \exp[-a(u - \theta)]}.$$

The following catalogue of parameters were employed in the respective simulation:

$$\begin{aligned} \tau_E &= 1.0, & \tau_I &= 2.4, & w_{EE} &= 12.0, & w_{EI} &= 8.0, \\ w_{IE} &= 10.0, & w_{II} &= 1.7, & P &= -1.7, & Q &= -2.6, \\ a &= 0.92, & \theta &= 0, & r &= 1.0. \end{aligned}$$

A.3. FitzHugh–Nagumo Excitable Dynamics

We consider the FitzHugh–Nagumo model (FitzHugh, 1961; Nagumo et al., 1962)

$$\dot{v} = v - \frac{v^3}{3} - w + I_{\text{ext}},$$
$$\dot{w} = \varepsilon\,(v + a - bw)\,,$$

where $v(t)$ represents the fast voltage-like variable and $w(t)$ is the slow recovery variable. We use the stable excitable regime with parameters $a = 0.7,\ b = 0.8,\ \varepsilon = 0.08,\ I_{\text{ext}} = 0.10$.

Appendix B. Proofs

B.1. Proof for Lemma 4

Proof For any $f \in \text{dom}(U) \subseteq \mathcal{H}_k$, the reproducing property gives

$$\langle Uf, k_x\rangle_{\mathcal{H}_k} = (Uf)(x) = f(\boldsymbol{F}(x)) = \langle f, k_{\boldsymbol{F}(x)}\rangle_{\mathcal{H}_k}$$
$$\implies |\langle Uf, k_x\rangle_{\mathcal{H}_k}| = \left|\langle f, k_{\boldsymbol{F}(x)}\rangle_{\mathcal{H}_k}\right| \le \|f\|_{\mathcal{H}_k}\,\|k_{\boldsymbol{F}(x)}\|_{\mathcal{H}_k}.$$

The second implication follows with the application of Cauchy–Bunyakovsky–Schwarz inequality. Thus, the functional $f \mapsto \langle Uf, k_x\rangle_{\mathcal{H}_k}$ is bounded with respect to the $\mathcal{H}_k$ norm on $\text{dom}(U)$. Hence $k_x \in \text{dom}(U^*)$. Since $\text{dom}(U)$ is dense in $\mathcal{H}_k$, the representing vector is unique and therefore $U^* k_x = k_{\boldsymbol{F}(x)}$. ■

B.2. Proof for Proposition 5

Proof For $h = \mathbf{\Phi}_{\mathcal{X}}\boldsymbol{c} = \sum_{i=1}^{M} c_i k_{x_i} \in V_{\mathcal{X}}$, the reproducing kernel Gram identities gives

$$\|h\|_{\mathcal{H}_k}^2 = \|\mathbf{\Phi}_{\mathcal{X}}\boldsymbol{c}\|_{\mathcal{H}_k}^2 = \boldsymbol{c}^*\mathbf{G}_{\mathcal{X}}\boldsymbol{c}.$$

By definition of $T_{\boldsymbol{F},\mathcal{X}}$, $T_{\boldsymbol{F},\mathcal{X}}h = \mathbf{\Phi}_{\mathcal{Y}}\boldsymbol{c}$, and hence $\|T_{\boldsymbol{F},\mathcal{X}}h\|_{\mathcal{H}_k}^2 = \boldsymbol{c}^*\mathbf{G}_{\mathcal{Y}}\boldsymbol{c}$. Since $\mathbf{G}_{\mathcal{X}} \succ 0$,

$$\|T_{\boldsymbol{F},\mathcal{X}}\|^2 = \sup_{\boldsymbol{c}\neq 0} \frac{\boldsymbol{c}^*\mathbf{G}_{\mathcal{Y}}\boldsymbol{c}}{\boldsymbol{c}^*\mathbf{G}_{\mathcal{X}}\boldsymbol{c}}.$$

Setting $b = \mathbf{G}_{\mathcal{X}}^{1/2}\boldsymbol{c}$ yields

$$\frac{\boldsymbol{c}^*\mathbf{G}_{\mathcal{Y}}\boldsymbol{c}}{\boldsymbol{c}^*\mathbf{G}_{\mathcal{X}}\boldsymbol{c}} = \frac{b^*\mathbf{G}_{\mathcal{X}}^{-1/2}\mathbf{G}_{\mathcal{Y}}\mathbf{G}_{\mathcal{X}}^{-1/2}b}{b^*b}.$$

Therefore, the Rayleigh–Ritz characterization gives $\|T_{\boldsymbol{F},\mathcal{X}}\|^2 = \lambda_{\max}\left(\mathbf{G}_{\mathcal{X}}^{-1/2}\mathbf{G}_{\mathcal{Y}}\mathbf{G}_{\mathcal{X}}^{-1/2}\right)$. ■

B.3. Proof for Proposition 9

Proof The first correlation bound follows from Cauchy–Bunyakovsky–Schwarz inequality in the $\mathbf{G}_\lambda$ inner-product defined in Definition 7, that is $|v^*\mathbf{G}_\lambda A_\lambda^t v| \leq \|v\|_{\mathbf{G}_\lambda}\|A_\lambda^t v\|_{\mathbf{G}_\lambda} \leq \|v\|_{\mathbf{G}_\lambda}^2\|A_\lambda\|_{\mathbf{G}_\lambda}^t$. The operator-norm identity is obtained by rewriting the $\mathbf{G}_\lambda$-induced norm in Euclidean coordinates:

$$\begin{aligned}\|A_\lambda\|_{\mathbf{G}_\lambda} &= \sup_{v\neq 0} \frac{\left\|\mathbf{G}_\lambda^{1/2}A_\lambda v\right\|_2}{\left\|\mathbf{G}_\lambda^{1/2}v\right\|_2} \\ &= \left\|\mathbf{G}_\lambda^{-1/2}\mathbf{A}_{\mathcal{X}}\mathbf{G}_\lambda^{-1/2}\right\|_2.\end{aligned} \tag{8}$$

Hence proved! ■

B.4. Proof for Theorem 10

Proof

1. Recall $\mathbf{A}_{\mathcal{X}} = \mathbf{\Phi}_{\mathcal{Y}}^*\mathbf{\Phi}_{\mathcal{X}}$, therefore $\widetilde{A}_\lambda = \mathbf{G}_\lambda^{-1/2}\mathbf{\Phi}_{\mathcal{Y}}^*\mathbf{\Phi}_{\mathcal{X}}\mathbf{G}_\lambda^{-1/2}$. Hence,

$$\begin{aligned}&\widetilde{A}_\lambda = \left(\mathbf{\Phi}_{\mathcal{Y}}\mathbf{G}_\lambda^{-1/2}\right)^*\left(\mathbf{\Phi}_{\mathcal{X}}\mathbf{G}_\lambda^{-1/2}\right), \\ \implies &\left\|\widetilde{A}_\lambda\right\|_2 \leq \left\|\mathbf{\Phi}_{\mathcal{Y}}\mathbf{G}_\lambda^{-1/2}\right\| \cdot \left\|\mathbf{\Phi}_{\mathcal{X}}\mathbf{G}_\lambda^{-1/2}\right\| \qquad \text{(by submultiplicativity)}.\end{aligned}$$

   Observe that

$$\begin{aligned}\left\|\mathbf{\Phi}_{\mathcal{Y}}\mathbf{G}_\lambda^{-1/2}\right\|^2 &= \left\|\mathbf{G}_\lambda^{-1/2}\mathbf{G}_{\mathcal{Y}}\mathbf{G}_\lambda^{-1/2}\right\|_2 = \chi_{k,\lambda}^2(\boldsymbol{F};\mathcal{X}), \text{ and} \\ \left\|\mathbf{\Phi}_{\mathcal{X}}\mathbf{G}_\lambda^{-1/2}\right\|^2 &= \left\|\mathbf{G}_\lambda^{-1/2}\mathbf{G}_{\mathcal{X}}\mathbf{G}_\lambda^{-1/2}\right\|_2.\end{aligned}$$

   Since $0 \preceq \mathbf{G}_{\mathcal{X}} \preceq \mathbf{G}_{\mathcal{X}} + \lambda I = \mathbf{G}_\lambda$, we have $\mathbf{G}_\lambda^{-1/2}\mathbf{G}_{\mathcal{X}}\mathbf{G}_\lambda^{-1/2} \preceq I$. Therefore, $\|\mathbf{\Phi}_{\mathcal{X}}\mathbf{G}_\lambda^{-1/2}\| \leq 1$. This immediately yields the desired result: $\left\|\widetilde{A}_\lambda\right\|_2 \leq \chi_{k,\lambda}(\boldsymbol{F};\mathcal{X})$.

2. Set $y = \mathbf{G}_\lambda^{1/2} v \implies v = \mathbf{G}_\lambda^{-1/2} y$ in (8). Then $\|v\|_{\mathbf{G}_\lambda}^2 = \left(\mathbf{G}_\lambda^{-1/2} y\right)^* \mathbf{G}_\lambda \left(\mathbf{G}_\lambda^{-1/2} y\right) = \|y\|_2^2$. Hence,

$$\begin{aligned}\left\|A_\lambda^t\right\|_{\mathbf{G}_\lambda} &= \sup_{y\neq 0} \frac{\left\|A_\lambda^t \mathbf{G}_\lambda^{-1/2} y\right\|_{\mathbf{G}_\lambda}}{\left\|\mathbf{G}_\lambda^{-1/2} y\right\|_{\mathbf{G}_\lambda}} \\ &= \sup_{y\neq 0} \frac{\left\|\mathbf{G}_\lambda^{1/2} A_\lambda^t \mathbf{G}_\lambda^{-1/2} y\right\|_2}{\|y\|_2} \\ &= \|\widetilde{A}_\lambda^t\|_2.\end{aligned}$$

3. Immediate consequence of previous two results.

4. Apply the spectral theorem on $\widetilde{A}_\lambda \in \mathbb{R}^{n\times n}$ to have $\widetilde{A}_\lambda = QDQ^*$ where Q is a unitary matrix and $D = \operatorname{diag}(\mu_1, \cdots, \mu_n)$ is a diagonal matrix. Therefore,

$$\left\|\widetilde{A}_\lambda^t\right\|_2 = \|D^t\|_2 = \max_j |\mu_j|^t = \left(\rho(\widetilde{A}_\lambda)\right)^t \implies \|A_\lambda^t\|_{\mathbf{G}_\lambda} = \left(\rho(\widetilde{A}_\lambda)\right)^t.$$

5. To prove this, we first recall the following result of the Kreiss matrix theorem (Trefethen and Embree, 2020) for a square matrix $\mathscr{A} \in \mathbb{C}^{n\times n}$ and Euler's number e:

$$\mathcal{K}(\mathscr{A}) \leq \sup_t \|\mathscr{A}^t\| \leq \mathrm{e}\cdot n \cdot \mathcal{K}(\mathscr{A}). \tag{9}$$

Let the assumption be true. Set $\mathcal{M} \coloneqq \sup_{t\geq 0} \|\widetilde{A}_\lambda^t\|_2$. For any $z \in \mathbb{C}$ with $|z| > 1$, the Neumann series expansion of the resolvent gives $(zI - \widetilde{A}_\lambda)^{-1} = \sum_t z^{-t-1} \widetilde{A}_\lambda^t$ which further implies

$$\left\|(zI - \widetilde{A}_\lambda)^{-1}\right\|_2 \leq \sum_t |z|^{-t-1} \left\|\widetilde{A}_\lambda^t\right\|_2 \leq \frac{\mathcal{M}}{|z|-1}.$$

Taking sup over the region $|z| > 1$ yields $\mathcal{K}(\widetilde{A}_\lambda) \leq \mathcal{M}$. Since $\mathcal{M} = \sup_{t\geq 0} \|A_\lambda^t\|_{\mathbf{G}_\lambda}$, we obtain $\mathcal{K}(\widetilde{A}_\lambda) \leq \sup_{t\geq 0} \|A_\lambda^t\|_{\mathbf{G}_\lambda}$. The extreme right side inequality up to a dimensional constant is the Kreiss matrix theorem $\sup_{t\geq 0} \|\widetilde{A}_\lambda^t\|_2 \leq \mathrm{e}\, n\, \mathcal{K}(\widetilde{A}_\lambda)$ from (9). Using $\|A_\lambda^t\|_{\mathbf{G}_\lambda} = \|\widetilde{A}_\lambda^t\|_2$, this respective part settles.

Finally, if $z \in \sigma_\varepsilon(\widetilde{A}_\lambda)$ with $|z| > 1$, then by definition $\|(zI - \widetilde{A}_\lambda)^{-1}\|_2 > \varepsilon^{-1}$. Hence, $(|z| - 1)\,\|(zI - \widetilde{A}_\lambda)^{-1}\|_2 > \frac{|z|-1}{\varepsilon}$. Take sup over all such z gives

$$\mathcal{K}(\widetilde{A}_\lambda) \geq \frac{r_\varepsilon(\widetilde{A}_\lambda) - 1}{\varepsilon},$$

where $r_\varepsilon(\widetilde{A}_\lambda)$ defined in (7). Combine the last inequality with $\mathcal{K}(\widetilde{A}_\lambda) \leq \sup_{t\geq 0} \|A_\lambda^t\|_{\mathbf{G}_\lambda}$; hence proved!

■

B.5. Proof for Corollary 11

Proof First combine Proposition 9 and Theorem 10(2,3) to have $\left|C_f^{(\lambda)}(t)\right| \leq \|v\|^2_{\mathbf{G}_\lambda} \left\|A_\lambda^t\right\|_{\mathbf{G}_\lambda} = \|v\|^2_{\mathbf{G}_\lambda} \left\|\widetilde{A}_\lambda^t\right\|_2 \leq \|v\|^2_{\mathbf{G}_\lambda} \chi_{k,\lambda}(\boldsymbol{F};\mathcal{X})^t$. Then divide this inequality by $C_f^{(\lambda)}(0) = \|v\|^2_{\mathbf{G}_\lambda}$ to have the following inequality:

$$\frac{\left|C_f^{(\lambda)}(t)\right|}{C_f^{(\lambda)}(0)} \leq \left\|A_\lambda^t\right\|_{\mathbf{G}_\lambda} = \left\|\tilde{A}_\lambda^t\right\|_2 \leq \chi_{k,\lambda}(\boldsymbol{F};\mathcal{X})^t.$$

Since $\frac{\left|C_f^{(\lambda)}(t)\right|}{C_f^{(\lambda)}(0)} \leq \sup \frac{\left|C_f^{(\lambda)}(t)\right|}{C_f^{(\lambda)}(0)} = \boldsymbol{w}(\tilde{A}_\lambda^t)$. From the general theory of numerical ranges of bounded linear operators (Gustafson and Rao, 1997), we get $\boldsymbol{w}(\tilde{A}_\lambda^t) \leq \left\|\tilde{A}_\lambda^t\right\|_2$. The final proof follows after necessary ordering. ■

Appendix C. Kernelized Regularization Sensitivity

Figure 4 shows that regularization acts as a systematic control on the transient response of the learned Koopman surrogates through KEDMD. Increasing λ lifts the small eigenvalues of the kernel Gram matrix $\mathbf{G}_\lambda$ and reduces the sensitivity associated with the inverse Gram geometry. Consequently, the strongest transient growth is progressively suppressed as regularization increases. This effect is highly representation dependent as the Gaussian RBF and rational-quadratic surrogates exhibit the largest variation with λ, whereas the Laplace surrogate remains comparatively insensitive over the tested range and Matérn-3/2 exhibits intermediate behavior.

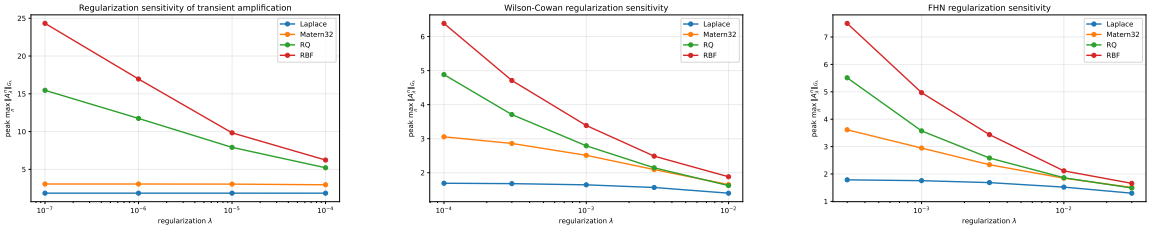


Figure 4: **Regularization sensitivity of representation-dependent transient amplification.** Peak induced operator growth, $\max_n \|A_\lambda^n\|_{\mathbf{G}_\lambda}$, is shown as a function of the Tikhonov regularization parameter λ.

Appendix D. Numerical Evaluation of the Kreiss constant

For each kernel and neural dynamical system, we estimate the discrete Kreiss constant using the following procedure:

1. Construct the regularized KEDMD operator $A_\lambda = (\mathbf{G}_\mathcal{X} + \lambda I)^{-1}\mathbf{A}_\mathcal{X}$.

2. Form the geometry-normalized operator $\widetilde{A}_\lambda = \mathbf{G}_\lambda^{1/2} A_\lambda \mathbf{G}_\lambda^{-1/2}$, $\mathbf{G}_\lambda = \mathbf{G}_\mathcal{X} + \lambda I$.
3. Define a Cartesian (complex) grid $\mathcal{G} = \{x_i + \mathrm{i}y_j\}$ in the complex plane and retain points satisfying $|z| > 1$.
4. For every retained grid point, compute $\sigma_{\min}(zI - \widetilde{A}_\lambda)$ and $R(z) = \dfrac{1}{\sigma_{\min}(zI - \widetilde{A}_\lambda)}$.
5. Evaluate $K(z) = (|z| - 1)R(z)$ and report $\widehat{\mathcal{K}} = \max_{z \in \mathcal{G},\, |z|>1} K(z)$.

Remark 12 *We highlight that $\widehat{\mathcal{K}}$ should not be interpreted as a direct prediction of the magnitude of the transient peak. Rather, it provides an independent resolvent-based diagnostic whose ordering tracks the independently computed finite-time amplification.*

To test whether this hierarchy depends on the numerical complex grid, we repeated the calculation on nested 31^2, 61^2, and 121^2 grids over the fixed domain $[-1.3, 1.3]^2$. Refining from 61^2 to 121^2 changes every estimate by at most approximately 1.1%, preserves the kernel ordering for all three systems, and places no maximizer on the outer computational boundary. The observed hierarchy therefore reflects a stable feature of the learned resolvent geometry rather than a particular grid discretization.

Table 3: Grid-resolution sensitivity of the discrete Kreiss diagnostic. Each normalized KEDMD surrogate is held fixed while only the Cartesian complex-grid resolution is varied over the common domain $\Re(z),\ \Im(z) \in [-1.3, 1.3]$.

System	**Kernel**	$\mathbf{31^2}$	$\mathbf{61^2}$	$\mathbf{121^2}$
Linear E–I	Laplace	1.370	1.370	1.371
	Matérn-3/2	1.618	1.676	1.676
	RQ	3.983	4.261	4.306
	RBF	5.527	5.773	5.773
Wilson–Cowan	Laplace	1.251	1.251	1.262
	Matérn-3/2	1.359	1.372	1.378
	RQ	1.646	1.658	1.662
	RBF	2.001	2.091	2.091
FitzHugh–Nagumo	Laplace	1.200	1.200	1.200
	Matérn-3/2	1.391	1.391	1.398
	RQ	1.486	1.486	1.489
	RBF	1.796	1.796	1.796

Appendix E. Pseudospectra for Neural Dynamical Systems

Figure 6 reveals that the kernel-dependent transient amplification observed in Figure 3 is accompanied by systematic changes in the resolvent geometry of the learned Koopman surrogates. Although normalized KEDMD operators have spectral radii close to unity, their pseudospectra differ substantially across representations. In particular, the high-resolvent

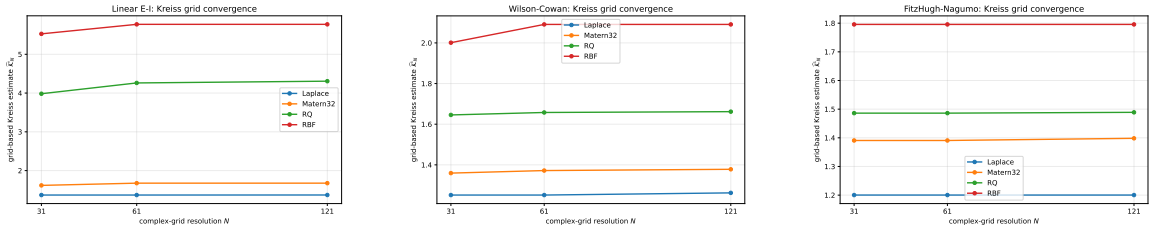


Figure 5: **Convergence of the grid-based Kreiss diagnostic under grid refinement.**

regions become increasingly concentrated toward the positive-real portion of the unit-circle boundary when moving from Laplace to Matérn-3/2, RQ, and RBF representations. The effect is strongest for the linear E–I benchmark, where the grid-based Kreiss estimate increases from 1.37 for Laplace to 5.77 for RBF, consistent with the corresponding increase in peak transient amplification from 1.85 to 16.96.

We note that the same qualitative ordering persists for the nonlinear Wilson–Cowan and FitzHugh–Nagumo systems. For Wilson–Cowan, $\widehat{\mathcal{K}}$ increases from 1.25 to 2.09, while for FitzHugh–Nagumo it increases from 1.20 to 1.80 between the Laplace and RBF representations. These system-dependent changes in resolvent sensitivity closely track the corresponding ordering of finite-time amplification.

Appendix F. Mechanistic view of Kernel Representation of Neural Dynamics

F.1. Observables used

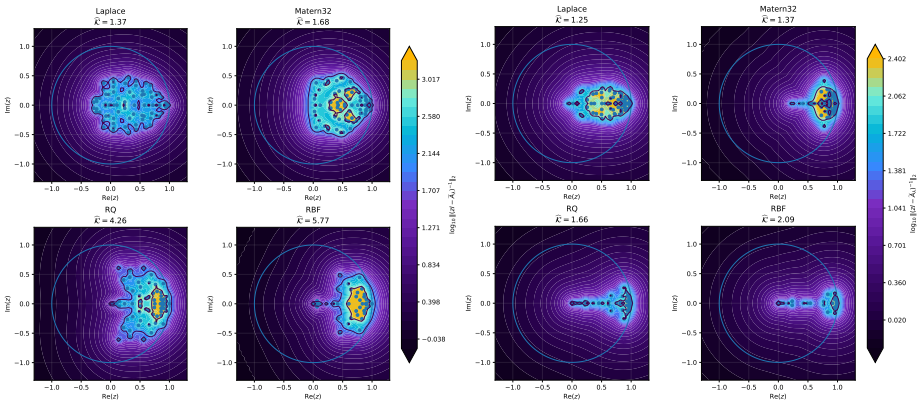
Table 4 reports the continuous observable used for the mechanistic view of the kernel representation for the neural dynamical system considered in this paper. The notation $E_\star$ and $v_\star$ denotes the respective equilibrium points for the respective neural dynamical systems.

Table 4: Observable functions used for the finite-time correlation analysis and Pareto-dominance experiments.

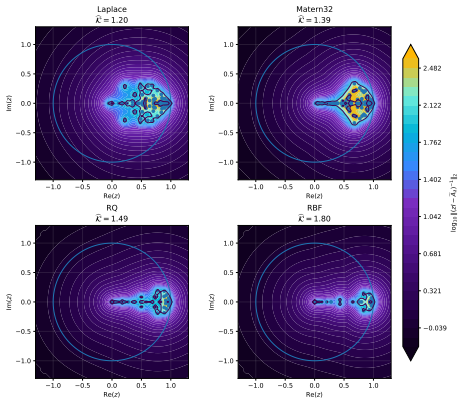
Neural system	**State vector**	**Observable f**
Stable linear E–I	$x = (\delta E, \delta I)$	$f(x) = \delta E$
Wilson–Cowan	$x = (E, I)$	$f(x) = E - E_\star$
FitzHugh–Nagumo	$x = (v, w)$	$f(x) = v - v_\star$

F.2. Mechanistic view

The following definition is for the dominance depth count that we used to plot for the contours inside the mechanistic view of the KEDMD surrogate for the neural dynamical systems.



(*a*) Linear E-I (*b*) Wilson-Cowan



(*c*) FitzHugh-Nagumo

Figure 6: **KEDMD surrogate pseudospectra of the Koopman operator for respective neural dynamical systems.** Color indicates the logarithm of the resolvent norm, $\log_{10} \left\| (zI - \widetilde{A}_\lambda)^{-1} \right\|_2$ where darker regions correspond to smaller resolvent amplification and therefore weaker sensitivity, whereas lighter regions indicate larger resolvent norms and stronger pseudospectral sensitivity.

Definition 13 (Dominance-Depth Count) *Let $\{(x_i, y_i)\}_{i=1}^{M}$ be $M < \infty$ points in the Cartesian plane. The dominance depth count $D : \mathbb{R} \times \mathbb{R} \to \mathbb{N}$ is a function given as follows:*

$$D(\xi, \eta) \coloneqq \#\{m : x_m \leq \xi, y_m \leq \eta, (x_m, y_m) \neq (\xi, \eta)\}$$

In our experiment, $D(\xi, \eta)$ counts how many sampled kernel–regularization pairs dominate the point (ξ, η) in the Pareto sense for a minimization problem.

In Figure 7, we provide a mechanistic perspective of kernel representation through a KEDMD surrogate for every choice of kernel considered on the stable E-I neural dynamical system; see Table 4 for the used observable. In particular, each point corresponds to one kernel–regularization configuration and is evaluated in the two-objective plane defined by

the peak KEDMD amplification and peak early-time observable correlation response, with lower values preferred along both axes.

The resulting organization in Figure 7 is strongly representation dependent where Laplace configurations occupy the lower-left region, Matérn-3/2 appears at intermediate levels, while RQ and especially RBF shift progressively toward larger transient gain and stronger observable response. The dominance-depth contours (see Definition 13 in Appendix F) quantify this ordering by indicating how many sampled configurations simultaneously outperform a given location in both objectives; therefore, increasing contour level toward the upper-right represents progressively more strongly dominated behavior rather than a probability density. In this experimental sweep, the Laplace kernel representation at regularization $\lambda = 10^{-4}$ emerges as the empirical Pareto optimum, indicating that it provides the most favorable joint balance of finite-time operator growth and early observable response among the tested configurations. The accompanying inset further links this behavior to the conditioning of the regularized kernel Gram matrix, showing that the configurations with larger transient amplification are also associated with poorer kernel-induced numerical geometry.

This figure along with Figures 8 and 9 supports the interpretation that finite-time reliability in the learned Koopman surrogate is not determined by the underlying neural dynamics alone, but is systematically shaped by the interaction between RKHS representation geometry and regularization.

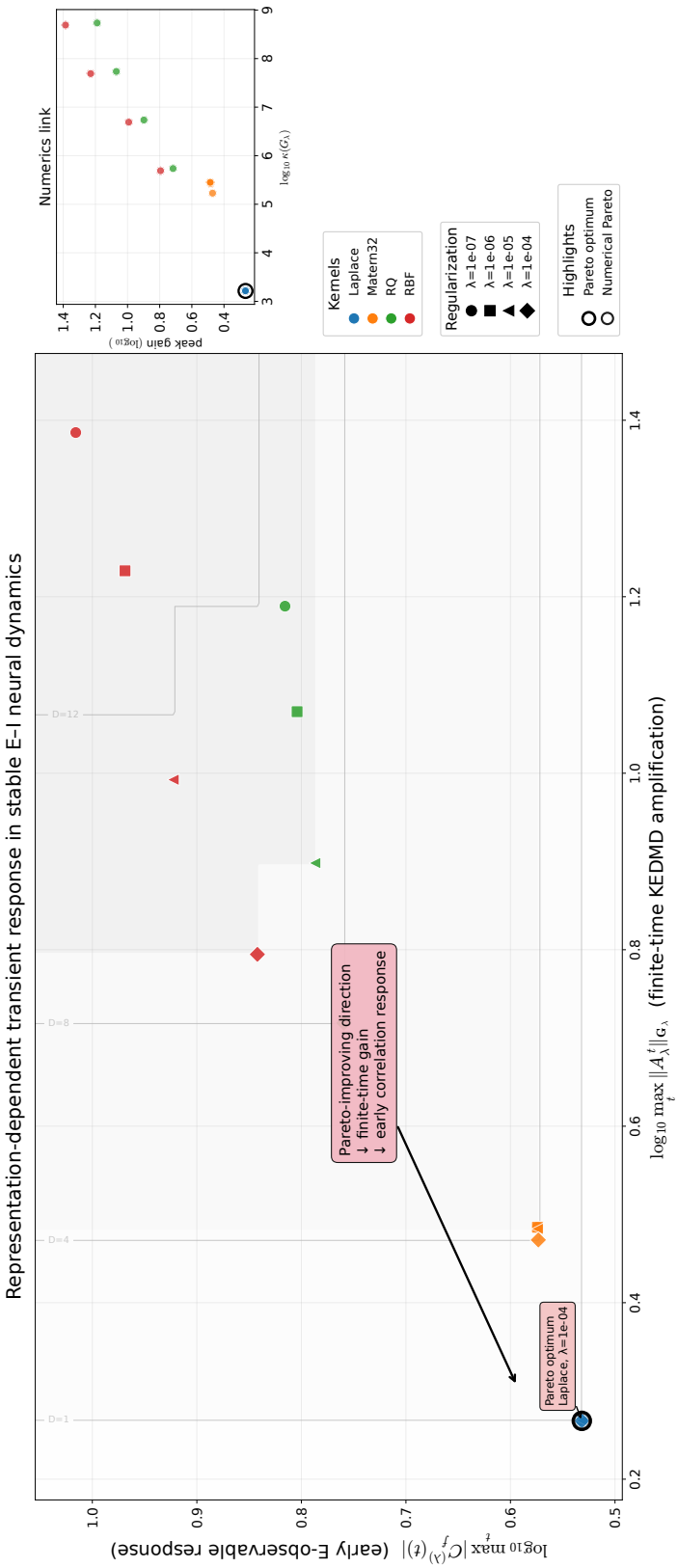


Figure 7: **Mechanistic perspective of KEDMD surrogates of the stable E–I.**

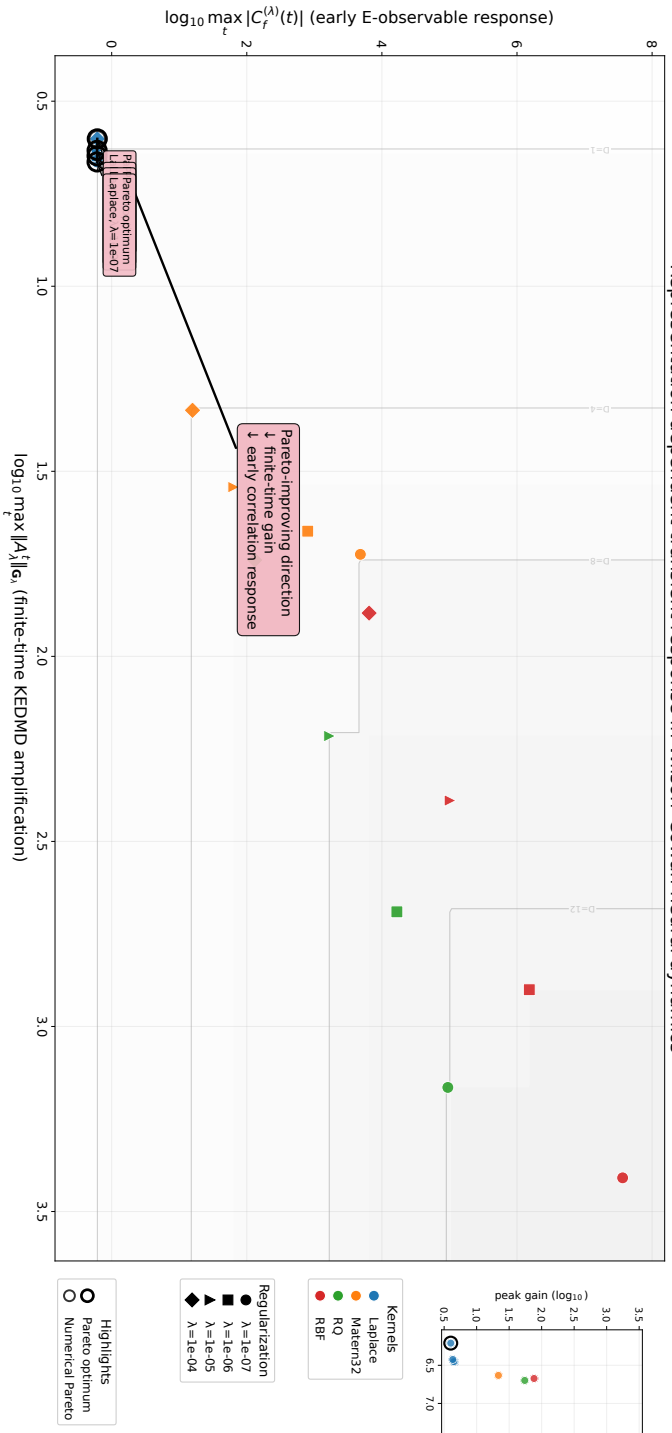


Figure 8: **Mechanistic perspective of KEDMD surrogates for Wilson–Cowan dynamics.**

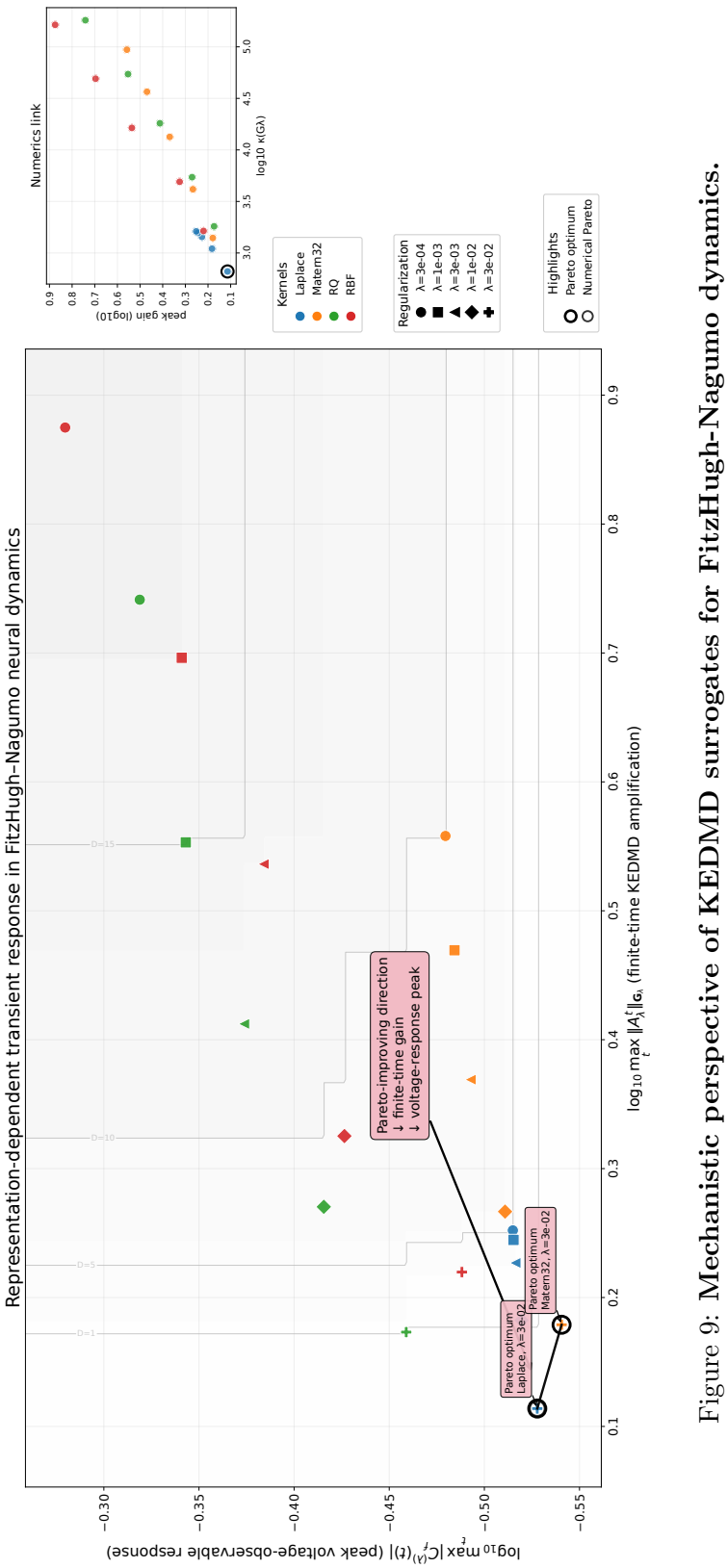


Figure 9: **Mechanistic perspective of KEDMD surrogates for FitzHugh-Nagumo dynamics.**